TELECOM CUSTOMER CHURN ANALYSIS

Executive Summary & Business Insights

Dataset	Records	Attributes
Telco Customer Churn	7,043 customers	21 columns

This report summarizes the exploratory analysis of telecom customer churn, highlights patterns observed across customer and service attributes, and translates those patterns into practical retention recommendations.

Executive Snapshot

26.54% of customers in the dataset are identified as churned, while **73.46%** remain customers. This indicates a meaningful retention challenge and provides a basis for examining which customer groups are more strongly associated with churn.

The analysis points to tenure, contract type, service subscriptions, senior-citizen status, internet-service category, and payment method as useful dimensions for further investigation. These are associations in the dataset, not proof that any single factor causes a customer to leave.

1. Project Overview

The objective of this project is to analyze customer churn in the telecom industry and identify customer and service characteristics associated with attrition. Customer churn refers to customers discontinuing the company's services. The analysis aims to understand customer behavior, identify higher-risk segments, and develop actionable recommendations that may support customer retention.

The dataset contains 7,043 customer records and 21 attributes covering customer demographics, tenure, contract information, subscribed services, billing details, and churn status.

2. Data Preparation and Cleaning

- **Data inspection:** Pandas functions such as `info()`, `describe()`, and `head()` were used to review the dataset structure, data types, and summary statistics.
- **Missing or blank values:** Blank entries in `TotalCharges` were identified. The analysis notes that these were replaced with 0 for customers with zero tenure and no recorded total charges.
- **Data type conversion:** `TotalCharges` was converted from object/string format to a numeric type for analysis.
- **Duplicate check:** Customer IDs were checked for duplicate records.
- **Data transformation:** The `SeniorCitizen` field was made easier to interpret by mapping 0 and 1 to No and Yes.

3. Exploratory Data Analysis and Key Findings

A. Overall Customer Churn

Approximately 26.54% of customers are marked as churned, compared with 73.46% who remain. In practical terms, this is slightly more than one in every four customers. The overall churn distribution establishes the scale of the retention challenge examined in the rest of the project.

B. Customer Demographics

The analysis compares churn across gender and senior-citizen groups. The gender comparison is useful for checking whether the distribution differs meaningfully between male and female customers. Senior citizens show a higher churn percentage than non-senior customers in the observations summarized for this project. This suggests that customer support and retention experiences for this segment may deserve closer review.

C. Customer Tenure

Tenure analysis indicates that customers with shorter relationships—particularly those in their earliest months—are more likely to churn. Customers who have stayed longer tend to show stronger retention. This pattern makes the early customer journey an important area for follow-up, including onboarding, service activation, and proactive support.

D. Contract Type

Month-to-month customers show considerably higher churn counts than customers on one-year or two-year contracts. Longer-term contract customers demonstrate comparatively stronger retention. The pattern may reflect differences in commitment, price plans, or customer experience; the chart alone does not establish which explanation is responsible.

E. Service Subscriptions

Countplots were used to compare churn across PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, and StreamingMovies. The charts show notable churn counts among customers without OnlineSecurity and TechSupport. Customers without OnlineBackup and DeviceProtection also contribute substantial churn counts. Fiber optic users represent the largest churn count among the internet-service categories in the visualization.

These are count comparisons. Because the number of customers in each category can differ, the counts should be supplemented with within-category churn rates before concluding that one group has a higher likelihood of churning.

F. Payment Method

Electronic check users account for a higher number of churned customers than users of other payment methods in the analysis summarized here. This makes the payment experience a useful area for further investigation. The count should be compared with the total number of customers using each payment method to establish the corresponding churn rate.

4. Major Business Insights

- **Early customer retention:** The first months of a customer relationship appear especially important. Early engagement and onboarding may help address issues before customers decide to leave.
- **Contract structure:** Month-to-month plans are associated with higher churn counts. Long-term plan incentives may be worth testing, while ensuring that the plans remain suitable for customer needs.
- **Service adoption:** Lower adoption of services such as OnlineSecurity and TechSupport is associated with notable churn counts. The business could explore whether clearer service education or relevant bundles improve perceived value.

- **Senior-citizen segment:** The higher churn percentage observed for senior citizens supports examining their service needs, support experience, and reasons for leaving.
- **Payment experience:** The electronic-check pattern merits further analysis of payment convenience, failed transactions, billing concerns, and customer preferences.
- **Fiber optic experience:** The high churn count among fiber optic customers warrants investigation of service quality, pricing, reliability, and support.

5. Business Recommendations

- **Strengthen onboarding:** Introduce welcome journeys, setup guidance, and early check-ins for newly acquired customers.
- **Test long-term contract offers:** Offer transparent discounts or loyalty benefits to month-to-month customers and measure whether conversion improves retention.
- **Promote relevant add-on services:** Explain the value of OnlineSecurity, TechSupport, OnlineBackup, and DeviceProtection through targeted recommendations rather than generic upselling.
- **Improve targeted support:** Review feedback and support journeys for senior citizens and other segments with elevated churn rates.
- **Review payment journeys:** Investigate the electronic-check experience and make alternative payment methods easy to understand and use.
- **Investigate fiber optic churn:** Examine complaints, outages, speed, pricing, and support resolution for fiber optic customers.
- **Monitor retention metrics:** Track churn rate by tenure, contract, payment method, internet service, and add-on subscriptions in a recurring dashboard.

6. Tools and Technologies

Tool	Purpose
Python	Data analysis and preprocessing
Pandas	Data cleaning, manipulation, and aggregation
NumPy	Numerical operations
Matplotlib	Data visualization
Seaborn	Statistical visualizations and countplots
Google Colab	Notebook development environment

7. Conclusion

The Telecom Customer Churn Analysis identifies several customer and service characteristics associated with churn, including tenure, contract type, service subscriptions, senior-citizen status, internet-service category, and payment method. The patterns suggest that early-tenure customers, month-to-month subscribers, and certain service and payment segments are important groups for further retention analysis.

The project demonstrates the use of Python for data cleaning, exploratory data analysis, visualization, and business insight generation. It provides a foundation for more targeted retention strategies and for further analytical work, such as percentage-based churn comparisons, customer segmentation, and predictive churn modeling.

8. Further Analysis to Strengthen the Project

The current visualizations include customer counts. To make comparisons between groups more reliable, calculate churn rate within each category—for example, churn rate by contract type, payment method, and internet-service category. A group with more customers can naturally have a higher churn count even when its churn percentage is not the highest.

Recommended next steps include a percentage-based churn dashboard, a comparison of monthly charges and total charges between churned and retained customers, and a customer-risk segmentation or predictive model. Any predictive model should be evaluated using appropriate train/test separation and metrics such as precision, recall, F1-score, and ROC-AUC.

> *Interpretation note: Findings in this report describe patterns observed in the exploratory analysis. They should be treated as associations, not causal conclusions. Where charts show counts rather than rates, percentage-based analysis is recommended before making group-level risk claims.*